TABER FISHER

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PROFESSIONAL SUMMARY

Senior AI/ML Engineer with 6+ years in geospatial intelligence and computer vision. Approaches problems by reading broadly and finding unexpected connections, like applying rendering techniques from game development to real time inference and pulling ideas from adjacent fields to figure out where they fit. Leads teams through technical depth, defining research directions, stress testing architectures, and building the tooling that makes the work move faster.

EXPERIENCE

Senior AI/ML Engineer • Lockheed Martin Autonomous Systems • Denver, CO *July 2024 – Present*

- Led a vision ML team researching model backbones and training strategies for panoptic segmentation; improved inference speeds 7 to 37 FPS across model variants while achieving SOTA Panoptic Quality (PQ) metrics; team won Q1 2026 innovation award
- Built automation pipelines for parallelized background training runs and a Streamlit app enabling project managers to interactively explore backbone trade-off curves
- Built an automated ML environment generation system using the Anthropic and OpenAI APIs to validate library requirements, generate Docker configurations, and self-heal on build failures before promoting to production

AI/ML Engineer • Lockheed Martin Space • Littleton, CO *January 2022 – July 2024*

- Led the Object-Level Change Detection (OLCD) project: a real-time geospatial intelligence system using React.js, vector databases, and advanced AI/ML algorithms with ActiveMQ for cross-domain event detection in overhead imagery
- Integration Lead for a novel sensor data fusion capability, designed and delivered full-stack integration including real-time backend processing (MySQL, Flask) and a flexible React-based UI/UX
- Adapted Facebook Omnivore and DeepMind PerceiverIO multi-modal architectures for geospatial recognition across EO and sensor fusion modalities; work contributed to a successful multimillion-dollar contract pitch
- Secured over $2M in internal R&D grants through proposal writing; recognized as 2023 Innovation Leader for creation of multiple trade secrets

Signals & Processing Engineer Asc. • Lockheed Martin Space • Littleton, CO *July 2020 – December 2021*

- Pioneered deep learning architectures for overhead computer vision applied to satellite and aerial imagery, improving inference speeds by 10x; contributed to the OLCD platform, which won a company-wide engineering innovation award
- Researched combining quantum computing with deep learning for fingerprinting and high-value object detection in overhead imagery

Software Development Intern • Lockheed Martin Space • Littleton, CO *May 2018 – December 2019*

- Supported the SAR Automatic Target Recognition (SAR ATR) team integrating neural network software with F-35 systems; built a GAN to synthesize SAR imagery for small-dataset augmentation
- Developed vehicle recognition software using TensorFlow trained on satellite and aerial imagery; authored a white paper proposing NLP-driven forecasting of emerging technology trends

Lead Software Architect • Rocketry@VT • Blacksburg, VA *September 2016 – September 2019*

- Developed a real-time aerial target tracking system in PyTorch for object identification during high-altitude flight; designed active drag system software for precise altitude targeting

AWARDS & PUBLICATIONS

- 2023 Innovation Leader — creation of multiple trade secrets and key geospatial AI projects *2023*
- LM Space Project Innovation Award — OLCD *2022*
- Publication: Object-Level Change Detection for Autonomous Sensemaking — SPIE Defense + Commercial Sensing *2022*
- Publication: Optimizing Deep Learning Classifier Performance for Low-to-Medium Resolution Overhead Imagery — SPIE Defense + Commercial Sensing *2021*
- Publication: Benchmarking Motor Imagery-Based BCI Classifiers Using EEG and iEEG Datasets — GlobalSIP *2019*

EDUCATION

M.S. Machine Learning — Georgia Institute of Technology *December 2025*

B.S. Computer Science, Magna Cum Laude — Virginia Tech *December 2019*

SKILLS

ML & AI: PyTorch, TensorFlow, Vision Transformers, CNNs, GANs, Multi-Modal Architectures, Semi-Supervised Learning, MARL, GNNs, Panoptic Segmentation

Geospatial: SAR / SAR ATR, Electro-Optical (EO), Multi-Spectral Imaging (MSI), Overhead Computer Vision

Languages & Tools: Python, JavaScript, C++, AWS, Docker, Linux, MySQL, MongoDB, Flask, FastAPI, React.js, TypeScript, Streamlit

Other: Blender, Godot, game development